DATA SCIENCE & ARTIFICIAL INTELLIGENCE

& CS/IT. – 2026/2027.

Linear Algebra

Eigen Vectors and Diagonalisation

Lecture No. 01



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Recap of previous lecture



Topic

1. C.H. Theorem
2. LU Decomposition

Topics to be Covered



Topic EIGEN VECTORS

For any square Mat $A_{n\times n}$ if $\boxed{AX=\lambda X} \Rightarrow (A-\lambda I)X=0$ — (1)

then X is called E-Vector ← it should be always Non Zero.

for eg E Vectors of $A=\begin{bmatrix}4&2\\2&4\end{bmatrix}$ are → $X_1=\begin{bmatrix}1\\1\end{bmatrix}$ for $\lambda_1=6$

→ $X_2=\begin{bmatrix}-1\\1\end{bmatrix}$ for $\lambda_2=2$

Cross check! $AX_1=\begin{bmatrix}4&2\\2&4\end{bmatrix}\begin{bmatrix}1\\1\end{bmatrix}=\begin{bmatrix}6\\6\end{bmatrix}=6\begin{bmatrix}1\\1\end{bmatrix}=6X_1$ Hence verified

Again $AX_2=\begin{bmatrix}4&2\\2&4\end{bmatrix}\begin{bmatrix}-1\\1\end{bmatrix}=\begin{bmatrix}-2\\2\end{bmatrix}=2\begin{bmatrix}-1\\1\end{bmatrix}=2X_2$ Hence verified

Procedure of finding EVectors →

for $A = \begin{bmatrix} 4 & 2 \\ 2 & 4 \end{bmatrix}$, C-Eq is

$$|A - \lambda I| = 0$$

$$\lambda^2 - (\text{Tr}\,A)\lambda + (|A|) = 0$$

$$\lambda^2 - 8\lambda + 12 = 0$$

$$\lambda = 6 \text{ & } 2$$

EVector for $(\lambda = 6)$ → Consider $AX = \lambda X$

$$(A - \lambda I)X = 0$$

$$(A - 6I)X = 0$$

$$\begin{bmatrix} (4-6) & 2 \\ 2 & (4-6) \end{bmatrix} X = 0$$

$$\begin{bmatrix} -2 & 2 \\ 2 & -2 \end{bmatrix} X = 0$$

$R_2 \to R_2 + R_1$, $\begin{bmatrix} -2 & 2 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}_{2\times 1}$

$$\begin{bmatrix} (-2x_1 + 2x_2) \\ 0 \end{bmatrix}_{2\times 1} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$\Rightarrow -2x_1 + 2x_2 = 0$ or $x_1 = x_2$

Let $x_2 = K$ then $x_1 = K$ where K = A constant

$$X = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} K \\ K \end{bmatrix} = K \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$ i.e for $\lambda = 6$, $X = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

$A=\begin{bmatrix}4&2\\2&4\end{bmatrix}$, E.Vector for $(\lambda=2)$:

Consider $AX=\lambda X$

$$(A-\lambda I)X=0$$

$$(A-2I)X=0$$

$$\begin{bmatrix}(4-2)&2\\2&(4-2)\end{bmatrix}X=0$$

$$\begin{bmatrix}2&2\\2&2\end{bmatrix}X=0$$

$R_2 \to R_2-R_1$

$$\begin{bmatrix}2&2\\0&0\end{bmatrix}\begin{bmatrix}x_1\\x_2\end{bmatrix}=\begin{bmatrix}0\\0\end{bmatrix}$$

$$\begin{bmatrix}(2x_1+2x_2)\\0\end{bmatrix}=\begin{bmatrix}0\\0\end{bmatrix}$$

$2x_1+2x_2=0 \Rightarrow x_1=-x_2$

Let $x_2=K$ then $x_1=-K$

$$\text{So } X=\begin{bmatrix}x_1\\x_2\end{bmatrix}=\begin{bmatrix}-K\\K\end{bmatrix}=K\begin{bmatrix}-1\\1\end{bmatrix}$$

i.e. for $\lambda=2$, $X=\begin{bmatrix}-1\\1\end{bmatrix}$

Conclusion:

$$A=\begin{bmatrix}4&2\\2&4\end{bmatrix}$$

$\lambda=6, X=\begin{bmatrix}1\\1\end{bmatrix}$

$\lambda=2, X=\begin{bmatrix}-1\\1\end{bmatrix}$

PODCAST (ANALYSIS) →

(1) $A=\begin{bmatrix}4&2\\2&4\end{bmatrix}$

$\lambda_1=6,\ X_1=\begin{bmatrix}1\\1\end{bmatrix}\rightarrow\begin{bmatrix}2\\2\end{bmatrix},\begin{bmatrix}3\\3\end{bmatrix},\begin{bmatrix}-3\\-3\end{bmatrix},\begin{bmatrix}-1\\-1\end{bmatrix},\begin{bmatrix}1/2\\1/2\end{bmatrix},\ldots\ldots\infty$ EV. exist

But all are LD on X_1

$\lambda_2=2,\ X_2=\begin{bmatrix}-1\\1\end{bmatrix}\rightarrow\begin{bmatrix}-1\\-1\end{bmatrix},\begin{bmatrix}3\\-3\end{bmatrix},\begin{bmatrix}-4/3\\4/3\end{bmatrix},\begin{bmatrix}-2\\2\end{bmatrix}\ldots\ldots\infty$ EVector exist

But all are LD on X_2

Here $X_1\cdot X_2=(1)(-1)+(1)(1)=0$

where X_1 & X_2 are L.I (By observation)

Cross check: $X=\begin{bmatrix}1/2\\1/2\end{bmatrix}$ then $AX=\begin{bmatrix}4&2\\2&4\end{bmatrix}\begin{bmatrix}1/2\\1/2\end{bmatrix}=\begin{bmatrix}3\\3\end{bmatrix}=6\begin{bmatrix}1/2\\1/2\end{bmatrix}=6X$

again taking $X=\begin{bmatrix}3\\-3\end{bmatrix}$ then $AX=\begin{bmatrix}4&2\\2&4\end{bmatrix}\begin{bmatrix}3\\-3\end{bmatrix}=\begin{bmatrix}6\\-6\end{bmatrix}=2\begin{bmatrix}3\\-3\end{bmatrix}=2X$

& Modal Mat $P=[X_1X_2]=\begin{bmatrix}1&-1\\1&1\end{bmatrix}$

(2) MODAL MATRIX: $P = [X_1 | X_2] = \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}$ $\because |P| \neq 0 \Rightarrow X_1 \& X_2$ are (LI)

Q. Find the E Values and E. Vectors of $A=\begin{bmatrix}1 & 2\\ 0 & 2\end{bmatrix}$

Sol: $\because$ A is U.T.M So E. Values are $\lambda = 1, 2$

E. Vector for $(\lambda=1) \longrightarrow$ $AX=\lambda X$

$$(A-\lambda I)X=0$$

$$(A-I)X=0$$

$$\begin{bmatrix}(1-1) & 2\\ 0 & (2-1)\end{bmatrix}X=0$$

$$\begin{bmatrix}0 & 2\\ 0 & 1\end{bmatrix}X=0$$

$R_2 \to R_2-\frac{1}{2}R_1$

$$\begin{bmatrix}0 & 2\\ 0 & 0\end{bmatrix}\begin{bmatrix}x_1\\ x_2\end{bmatrix}=\begin{bmatrix}0\\ 0\end{bmatrix}$$

$$\begin{bmatrix}2x_2\\ 0\end{bmatrix}=\begin{bmatrix}0\\ 0\end{bmatrix} \Rightarrow 2x_2=0 \Rightarrow \boxed{x_2=0}$$

Let $x_1=K$ then

$$X=\begin{bmatrix}x_1\\ x_2\end{bmatrix}=\begin{bmatrix}K\\ 0\end{bmatrix}=K\begin{bmatrix}1\\ 0\end{bmatrix}$$

Hence E. Vector for $\lambda=1$, $X=\begin{bmatrix}1\\ 0\end{bmatrix}$

E.Vector for $(\lambda=2)$ $\rightarrow$ $A=\begin{bmatrix}1 & 2\\ 0 & 2\end{bmatrix}$

Consider $AX=\lambda X$

$$(A-\lambda I)X=0$$

$$(A-2I)X=0$$

$$\begin{bmatrix}(1-2) & 2\\ 0 & (2-2)\end{bmatrix}X=0$$

$$\begin{bmatrix}-1 & 2\\ 0 & 0\end{bmatrix}\begin{bmatrix}x_1\\ x_2\end{bmatrix}=\begin{bmatrix}0\\ 0\end{bmatrix}$$

$$\begin{bmatrix}(-x_1+2x_2)\\ 0\end{bmatrix}=\begin{bmatrix}0\\ 0\end{bmatrix}$$

$-x_1+2x_2=0 \Rightarrow x_1=2x_2$

Let $x_2=K$ then $x_1=2K$

so $X=\begin{bmatrix}x_1\\ x_2\end{bmatrix}=\begin{bmatrix}2K\\ K\end{bmatrix}=K\begin{bmatrix}2\\ 1\end{bmatrix}$

is E.vector for $\lambda=2$, $X=\begin{bmatrix}2\\ 1\end{bmatrix}$

for $A=\begin{bmatrix}1 & 2\\ 0 & 2\end{bmatrix}$ $\rightarrow$ $\lambda=1$, $X=\begin{bmatrix}1\\ 0\end{bmatrix}$

$\lambda=2$, $X=\begin{bmatrix}2\\ 1\end{bmatrix}$

PROPERTIES of EVECTORS →

① we are free to Multiply or divide with any Constant in Case of E.Vectors.

$\because AX = \lambda X \Rightarrow \boxed{(A-\lambda I)X = O} \Rightarrow K(A-\lambda I)X = KO \Rightarrow \boxed{(A-\lambda I)(KX) = O}$

① — Non Zero E Vector

Non Zero $\&$ ∞ $\&$

② — Non Zero E Vector

i.e whenever X is an E.Vector $\Leftrightarrow$ (KX) will also be an E.Vector

② E Vectors for different E Values are LI

③ If E Value Repeats (then Headache starts) then Corresponding E Vectors may be LI or may be LD.

(4) **MODAL MATRIX:** MATRIX formed by E.Vectors is called Modal Matrix.

Let $A_{3\times3}$ & it's EVectors are X_1, X_2, X_3

then Modal Mat $P = [X_1\ X_2\ X_3]$

if $|P| \neq 0 \Rightarrow$ EVectors are LI & if $|P| = 0 \Rightarrow$ E.Vectors are (LD)

(5) E.Vectors Corresponding to different E.Values of Symmetric Mat are orthogonal

(2015) Consider Symm Mat $A_{3\times3}$ s.t it's E.Values are a & b & Corresponding EVectors are $\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$ & $\begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}$ then for $a \neq b$ the value of $(x_1y_1 + x_2y_2 + x_3y_3) = ?$

$X =$ $Y =$

By above prop: $X \cdot Y = 0 \Rightarrow x_1y_1 + x_2y_2 + x_3y_3 = 0$

⑥ Algebraic Multiplicity (A.M) = Number of times λ Repeats is called AM of λ

Geometric Multiplicity (G.M) = Counting of LI E.Vectors for λ is called GM of λ

Shortcut of finding GM → Let λ is the Repeated E.Value then

$$\text{GM of } \lambda = \text{order} - \rho(A - \lambda I)$$

ie Number of L.I E.Vectors for λ = order $- \rho(A - \lambda I)$

Q: Find the E.Values & E.Vectors of $A=\begin{bmatrix}2 & 1\\ 0 & 2\end{bmatrix}$

Sol: $\because A$ is U.T.M $\therefore \lambda = 2, 2$ (Repeats)

E.Vector for ($\lambda=2$): Consider $AX=\lambda X$

$$(A-\lambda I)X=0$$

$$(A-2I)X=0$$

$$\begin{bmatrix}0 & 1\\ 0 & 0\end{bmatrix}\begin{bmatrix}x_1\\ x_2\end{bmatrix}=\begin{bmatrix}0\\ 0\end{bmatrix}$$

$$\begin{bmatrix}x_2\\ 0\end{bmatrix}=\begin{bmatrix}0\\ 0\end{bmatrix}$$

$x_2=0$ & let $x_1=K$

Hence $X=\begin{bmatrix}x_1\\ x_2\end{bmatrix}=\begin{bmatrix}K\\ 0\end{bmatrix}=K\begin{bmatrix}1\\ 0\end{bmatrix}$

i.e for $\lambda=2$, $X=\begin{bmatrix}1\\ 0\end{bmatrix}$

$X=\begin{bmatrix}1\\ 0\end{bmatrix} \longmapsto \begin{bmatrix}2\\ 0\end{bmatrix}, \begin{bmatrix}-3\\ 0\end{bmatrix}, \begin{bmatrix}4/5\\ 0\end{bmatrix}, \ldots\ldots$

But all are LD on X

For $A=\begin{bmatrix}2 & 1\\ 0 & 2\end{bmatrix}$ & $\lambda=2$, $X=\begin{bmatrix}1\\ 0\end{bmatrix}$.

($\lambda=2,2$)

AMof($\lambda=2$) = Two & GMof($\lambda=2$) = one

Q: Find the E-Vectors of $A=\begin{bmatrix}0&0&\alpha\\0&0&0\\0&0&0\end{bmatrix}$, $\alpha\neq 0$

Sol: $\because$ A is U.T.M $\therefore \lambda = 0,0,0$

E-Vector for $(\lambda=0)\rightarrow$

$$AX=\lambda X$$
$$(A-\lambda I)X=0$$
$$(A-0I)X=0$$
$$AX=0$$
$$\begin{bmatrix}0&0&\alpha\\0&0&0\\0&0&0\end{bmatrix}_{3\times 3}\begin{bmatrix}x_1\\x_2\\x_3\end{bmatrix}=\begin{bmatrix}0\\0\\0\end{bmatrix}$$

$$\begin{bmatrix}\alpha x_3\\0\\0\end{bmatrix}=\begin{bmatrix}0\\0\\0\end{bmatrix}\Rightarrow \alpha x_3=0$$
or $x_3=0$

Let $x_2=\lambda$, $x_1=\mu$

$$X=\begin{bmatrix}x_1\\x_2\\x_3\end{bmatrix}=\begin{bmatrix}\mu\\\lambda\\0\end{bmatrix}=\begin{bmatrix}\mu\\0\\0\end{bmatrix}+\begin{bmatrix}0\\\lambda\\0\end{bmatrix}$$
$$X=\mu\begin{bmatrix}1\\0\\0\end{bmatrix}+\lambda\begin{bmatrix}0\\1\\0\end{bmatrix}$$

i.e for $\lambda=0$ $\rightarrow$ $X_1=\begin{bmatrix}1\\0\\0\end{bmatrix}$, $X_2=\begin{bmatrix}0\\1\\0\end{bmatrix}$

if for $A=\begin{bmatrix}0&0&\alpha\\0&0&0\\0&0&0\end{bmatrix}_{3\times3}$ & for $\lambda=0$ $(\lambda=0,0,0)$

$\rightarrow X_1=\begin{bmatrix}1\\0\\0\end{bmatrix}\rightarrow\begin{bmatrix}-2\\0\\0\end{bmatrix},\begin{bmatrix}3\\0\\0\end{bmatrix},\begin{bmatrix}-1\\0\\0\end{bmatrix}\cdots\cdots\infty$ E.Vectors

But all are LD on X_1

$\rightarrow X_2=\begin{bmatrix}0\\1\\0\end{bmatrix}\rightarrow\begin{bmatrix}0\\-2\\0\end{bmatrix},\begin{bmatrix}0\\5\\0\end{bmatrix}\cdots\infty$ E.Vectors

But all are LD on X_2

where X_1 & X_2 are (LI)

Note:

① AMof $(\lambda=0)$ = three & GMof $(\lambda=0)$ = two

② GMof $(\lambda=0)$ = order $-\rho(A-\lambda I)$

$= 3-\rho(A-0I)=3-\rho(A)=3-1=2$

Q: Find E.Vectors of $A=\begin{bmatrix}2&1&1\\1&2&1\\0&0&1\end{bmatrix}$

Sol: C.Eq is $|A-\lambda I|=0 \Rightarrow \lambda=3,1,1$

Evector for $(\lambda=3) \longrightarrow AX=\lambda X$

$$(A-\lambda I)X=0$$

$$(A-3I)X=0$$

$$\begin{bmatrix}-1&1&1\\1&-1&1\\0&0&-2\end{bmatrix}X=0$$

R_2+R_1

$$\begin{bmatrix}-1&1&1\\0&0&2\\0&0&-2\end{bmatrix}X=0$$

$R_3 \to R_3+R_2$,

$$\begin{bmatrix}-1&1&1\\0&0&2\\0&0&0\end{bmatrix}\begin{bmatrix}x_1\\x_2\\x_3\end{bmatrix}=\begin{bmatrix}0\\0\\0\end{bmatrix}$$

$$\begin{bmatrix}(-x_1+x_2+x_3)\\2x_3\\0\end{bmatrix}=\begin{bmatrix}0\\0\\0\end{bmatrix}$$

$$\left.\begin{aligned}-x_1+x_2+x_3=0\\x_3=0\end{aligned}\right\} \quad x_1=x_2=k \text{ (let)}$$

$$4x_3=0$$

$$X=\begin{bmatrix}x_1\\x_2\\x_3\end{bmatrix}=\begin{bmatrix}k\\k\\0\end{bmatrix}=k\begin{bmatrix}1\\1\\0\end{bmatrix}$$

ie for $\lambda=3$, $X=\begin{bmatrix}1\\1\\0\end{bmatrix}$

E.Vector for $(\lambda=1)$ $\rightarrow$ $A=\begin{bmatrix}2&1&1\\1&2&1\\0&0&1\end{bmatrix}$

Consider $AX=\lambda X$

$$(A-\lambda I)X=0$$

$$(A-I)X=0$$

$$\begin{bmatrix}1&1&1\\1&1&1\\0&0&0\end{bmatrix}X=0$$

R_2-R_1,

$$\begin{bmatrix}1&1&1\\0&0&0\\0&0&0\end{bmatrix}\begin{pmatrix}x_1\\x_2\\x_3\end{pmatrix}=\begin{pmatrix}0\\0\\0\end{pmatrix}$$

$$\begin{bmatrix}(x_1+x_2+x_3)\\0\\0\end{bmatrix}=\begin{bmatrix}0\\0\\0\end{bmatrix}$$

$x_1+x_2+x_3=0$

Let $x_3=\alpha$, $x_2=\beta$, $x_1=-(\alpha+\beta)$

$$X=\begin{bmatrix}x_1\\x_2\\x_3\end{bmatrix}=\begin{bmatrix}-(\alpha+\beta)\\\beta\\\alpha\end{bmatrix}=\begin{bmatrix}-\alpha\\0\\\alpha\end{bmatrix}+\begin{bmatrix}-\beta\\\beta\\0\end{bmatrix}$$

$$X=\alpha\begin{bmatrix}-1\\0\\1\end{bmatrix}+\beta\begin{bmatrix}-1\\1\\0\end{bmatrix}$$

for $(\lambda=1)$ $\rightarrow$ $X=\begin{bmatrix}-1\\0\\1\end{bmatrix}$, $X=\begin{bmatrix}-1\\1\\0\end{bmatrix}$

ie for

$A=\begin{bmatrix}2&1&1\\1&2&1\\0&0&1\end{bmatrix}$

$(\lambda=3,1,1)$

$\lambda=3$, $X_1=\begin{bmatrix}1\\1\\0\end{bmatrix}\rightarrow\begin{bmatrix}3\\3\\0\end{bmatrix},\begin{bmatrix}4/5\\4/5\\0\end{bmatrix}$ ---- ∞ Evectors exist

$\lambda=1$, $X_2=\begin{bmatrix}-1\\1\\0\end{bmatrix}\rightarrow\begin{bmatrix}-2\\2\\0\end{bmatrix},\begin{bmatrix}-1\\-1\\0\end{bmatrix}$ ---- " " "

$X_3=\begin{bmatrix}-1\\0\\1\end{bmatrix}\rightarrow\begin{bmatrix}-4\\0\\4\end{bmatrix},\begin{bmatrix}5\\0\\-5\end{bmatrix}$ ---- " " "

where X_1, X_2, X_3 are (LI) $\therefore$ Modal Mat P is Non Singular

where Modal Mat $P=[X_1\,X_2\,X_3]=\begin{bmatrix}1&-1&-1\\1&1&0\\0&0&1\end{bmatrix}$, $|P|\neq 0$

(2) AM of 3 = one, GM of 3 = one

AM of 1 = two, GM of 1 = two

Note GM of $(\lambda=1)$ by Shortcut Method :-

$$GM \text{ of } (\lambda=1) = \text{order} - \rho(A-\lambda I)$$

$$= 3 - \rho(A - 1.I)$$

$$A=\begin{bmatrix}2&1&1\\1&2&1\\0&0&1\end{bmatrix}. \qquad = 3-\rho\begin{bmatrix}1&1&1\\1&1&1\\0&0&0\end{bmatrix} = 3-1=2$$

Note: It is **Not digestable** that for $\lambda=1$, two different E vectors exist ??

for $\lambda=1$:

$X_1=\begin{bmatrix}-1\\0\\1\end{bmatrix}$ i.e $AX_1=1.X_1$

$X_2=\begin{bmatrix}-1\\1\\0\end{bmatrix}$ i.e $AX_2=1.X_2$

S.M + M.H = G.Slaun.

S.M + S.W = G.Slaun.

Note :- How to Break E. Vector

Let $X = \begin{bmatrix} -(\alpha+\beta) \\ \beta \\ \alpha \end{bmatrix}$

(M-I) By Common Sense ; $X = \begin{bmatrix} -\alpha \\ 0 \\ \alpha \end{bmatrix} + \begin{bmatrix} -\beta \\ \beta \\ 0 \end{bmatrix}$

(M-II) $X = \begin{bmatrix} -(\alpha+\beta) \\ \beta \\ \alpha \end{bmatrix}$

Put $\alpha = 0$ → $\begin{bmatrix} -\beta \\ \beta \\ 0 \end{bmatrix} = \beta \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}$

Put $\beta = 0$ → $\begin{bmatrix} -\alpha \\ 0 \\ \alpha \end{bmatrix} = \alpha \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$

Note :- We Can not put $\alpha = \beta = 0$ Simultaneously ∴ In that Situation E. Vector $X = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$ Which is not Possible.

THANK - YOU

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